# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 \| SUB-STRAND 1.2: INDICES | |
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| **Student Name** | ________________________________________ |
| **Class** | ________________________________________ |
| **Date** | ________________________________________ |

## INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Use it to write a complete, well-structured answer to the driving question. Read each section prompt carefully, study the exemplar response, and then write your own answer in your exercise book or on the platform. Your goal is to show that you truly understand — not just memorise — why indices exist and how their laws work. Write in full sentences, use mathematical notation correctly, and connect every idea back to real examples from Kenya and the world around you.

## Section 1: Why Do We Need Indices? (The Problem with Big and Small Numbers)

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| **Explain why scientists, economists, and government planners use index notation instead of writing numbers out in full. Use the Kenya 2019 Census figure as your starting point. What problem does index notation solve, and what is the general meaning of the expression $a^n$?** | When Kenya conducted its 2019 National Population and Housing Census, the total population recorded was 47,564,296 people. Written out in full like that, the number is already difficult to read — imagine trying to compare it with the population of Africa (about 1,340,000,000) or the number of bacteria in a teaspoon of soil (roughly 1,000,000,000,000). Each extra zero is a chance to make a mistake, and multiplying or dividing such numbers by hand becomes a nightmare.<br><br>Index notation solves this problem by using repeated multiplication in a compact form. Instead of writing 10 × 10 × 10 × 10 × 10 × 10 × 10, you write $10^7$ — the base 10 tells you what is being multiplied, and the index (or exponent) 7 tells you how many times. In general, the expression $a^n$ means 'multiply the base a by itself n times': $a^n = a \times a \times a \times \ldots \times a$ (n times).<br><br>So 47,564,296 can be approximated as $4.8 \times 10^7$. The tiny superscript 7 carries enormous information in a single symbol. This is why a Kenya Meteorological Department scientist recording rainfall in microlitres, a CBK economist tracking national debt in billions of shillings, or a Rift Valley maize farmer calculating fertiliser concentrations all reach for index notation — it makes huge and tiny quantities manageable, comparable, and far less likely to be misread. |

## Section 2: The Multiplication and Division Laws of Indices

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| **State and explain the multiplication law ($a^m \times a^n =$** | The Multiplication Law — $a^m \times a^n = a^{m+n}$ |

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| **$a^{m+n}$) and the division law ($a^m \div a^n = a^{m-n}$). Show clearly WHY each law works by expanding the repeated multiplication. Give a real-life example for each law.** | This law says: when you multiply two powers that have the SAME base, you simply ADD the indices. But why? Expand both powers to see:<br><br>$a^3 \times a^4 = (a \times a \times a) \times (a \times a \times a \times a) = a^7$<br><br>You are not doing anything magical — you are just counting how many times a appears as a factor altogether: 3 + 4 = 7. The base stays the same; the indices add.<br><br>Real-life example: A scientist at the Kenya Medical Research Institute (KEMRI) is studying a bacterial culture. At hour zero there are $10^3$ bacteria. Every hour the population multiplies by $10^2$. After one such hour the population is $10^3 \times 10^2 = 10^5 = 100{,}000$ bacteria. Adding indices (3 + 2 = 5) gives the answer instantly — no long multiplication needed.<br><br>The Division Law — $a^m \div a^n = a^{m-n}$<br><br>This law says: when you DIVIDE two powers with the same base, SUBTRACT the indices. Again, expand to see why:<br><br>$a^5 \div a^2 = (a \times a \times a \times a \times a) \div (a \times a) = a \times a \times a = a^3$<br><br>Two factors of a in the numerator cancel with two in the denominator, leaving 5 − 2 = 3 factors.<br><br>Real-life example: A Kericho tea estate stores $10^6$ litres of water in its reservoir. Each irrigation cycle uses $10^4$ litres. The number of full cycles possible is $10^6 \div 10^4 = 10^2 = 100$ cycles. Subtracting indices (6 − 4 = 2) gives the answer in one step. |

| Section 3: The Power Law and the Zero Index | |
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| **Explain the power of a power law — $(a^m)^n = a^{mn}$ — and prove why any non-zero number raised to the power zero equals exactly 1. Many students find the zero index surprising; your explanation should show it is not a coincidence or a rule someone made up, but a logical consequence of the division law.** | The Power of a Power Law — $(a^m)^n = a^{mn}$<br><br>When a power is itself raised to another power, multiply the indices. Here is why:<br><br>$(a^2)^3 = a^2 \times a^2 \times a^2 = (a \times a) \times (a \times a) \times (a \times a) = a^6$<br><br>You multiplied $a^2$ three times, so you added the index 2 exactly 3 times: 2 + 2 + 2 = 2 × 3 = 6. Repeated addition of the same index is just multiplication, so $(a^m)^n = a^{mn}$.<br><br>Real-life example: Kenya's national grid transmits electrical power. An engineer expresses a voltage as $(10^3)^2$ volts. Using the power law: $(10^3)^2 = 10^6 = 1{,}000{,}000$ volts — calculated instantly.<br><br>Why Does $a^0 = 1$? (The Zero Index) |

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| | This is the result that surprises most learners. It seems strange that $10^0 = 1$, or that $47^0 = 1$. But it follows directly from the division law — it is not a definition plucked from thin air.<br><br>Consider: $a^n \div a^n$. Any number divided by itself equals 1, so $a^n \div a^n = 1$.<br><br>Now apply the division law to the same expression: $a^n \div a^n = a^{n-n} = a^0$.<br><br>Both results describe exactly the same calculation. Therefore $a^0$ must equal 1.<br><br>Example: $10^5 \div 10^5 = 100{,}000 \div 100{,}000 = 1$, and by the division law $= 10^{5-5} = 10^0$. So $10^0 = 1$. This works for any non-zero base. A census planner dividing any population figure by itself — say, checking that a ratio equals one — is using this exact logic every time they simplify such expressions. |

| Section 4: Negative and Fractional Indices | |
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| **Explain what a negative index means ($a^{-n} = 1/a^n$) and what a fractional index means ($a^{(1/n)} = \sqrt[n]{a}$). Use the division law to show WHY a negative index produces a reciprocal, and give one real-life Kenyan context for each.** | Negative Indices — $a^{-n} = 1/a^n$<br><br>A negative index does NOT mean a negative number. It means a reciprocal. Once more, the division law reveals why.<br><br>Consider $a^2 \div a^5$:<br>– By division law: $a^2 \div a^5 = a^{2-5} = a^{-3}$<br>– By expanding: $(a \times a) \div (a \times a \times a \times a \times a) = 1/a^3$<br><br>Both expressions describe the same division, so $a^{-3} = 1/a^3$. In general, $a^{-n} = 1/a^n$.<br><br>Real-life example: In chemistry class at a Nairobi secondary school, the concentration of hydrochloric acid used in a titration experiment might be $10^{-2}$ mol/L (0.01 mol/L). The negative index tells you immediately that the quantity is a tiny fraction — much easier than writing many zeros after the decimal point. A student misreading $10^{-2}$ as a negative number would make a dangerous error in the lab.<br><br>Fractional Indices — $a^{(1/n)} = \sqrt[n]{a}$<br><br>A fractional index represents a root. To understand why, use the power law:<br><br>$(a^{(1/2)})^2 = a^{(1/2 \times 2)} = a^1 = a$<br><br>So $a^{(1/2)}$, when squared, gives a. That is exactly what a square root does! Therefore $a^{(1/2)} = \sqrt{a}$. Similarly, $a^{(1/3)} = \sqrt[3]{a}$ (the cube root), because $(a^{(1/3)})^3 = a$. |

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| | Real-life example: A farmer in Nakuru wants to know the side length of a square plot that has an area of $10^4$ m². The side = $(10^4)^{(1/2)} = 10^{(4\times½)} = 10^2 = 100$ m. Using fractional index notation, the calculation is clean and quick — no guessing needed. |

## Section 5: Answering the Driving Question — Putting It All Together

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| **Now answer the driving question directly and in full: 'Why do scientists, economists and farmers use small superscript numbers to write huge or tiny quantities — and what rules make those compact symbols so powerful for computation?' Your answer must mention: (a) the purpose of index notation, (b) at least three laws of indices and why they work, (c) at least two different Kenyan real-life contexts, and (d) a reflection on what would be lost if we could NOT use index notation.** | Scientists, economists, and farmers use index notation — those small superscript numbers — because it is the most efficient, accurate, and logical way to handle the enormous range of quantities that appear in the real world. From Kenya's census population of approximately $4.8 \times 10^7$ people, to the $10^{-6}$ metre width of a red blood cell studied at Kenyatta University, to the $10^{12}$ shillings in a national budget projection, these quantities span ranges that ordinary decimal notation makes unwieldy and error-prone.<br><br>The compact power $a^n$ carries two pieces of information: the base a (what is being multiplied) and the index n (how many times). This structure is not just convenient — it comes with a set of laws that make arithmetic with these numbers surprisingly elegant:<br><br>– The Multiplication Law ($a^m \times a^n = a^{m+n}$) works because you are simply counting all the repeated factors together. A KEMRI scientist combining two bacterial growth stages multiplies populations by adding indices — no long multiplication required.<br><br>– The Division Law ($a^m \div a^n = a^{m-n}$) works because common factors cancel. A Kericho tea farmer calculating how many irrigation cycles fit into a water reserve subtracts one index from another and has the answer.<br><br>– The Zero Index ($a^0 = 1$) is not a strange exception — it is forced by the division law itself, because any non-zero number divided by itself must equal 1.<br><br>– The Negative Index ($a^{-n} = 1/a^n$) extends the same logic to cases where the denominator has more factors, turning the expression into a reciprocal — essential for expressing tiny concentrations in chemistry or microbiology.<br><br>– The Fractional Index ($a^{(1/n)} = \sqrt[n]{a}$) connects powers to roots through the power-of-a-power law, allowing a Nakuru farmer to calculate the dimensions of a field from its area in a single step.<br><br>Without index notation, every one of these calculations would require writing out chains of tens, hundreds, or thousands of multiplied factors. Errors would multiply as fast as bacteria. Comparisons across very large and very small quantities — the kind a government planner makes when comparing population density in Nairobi versus a rural Turkana county — would become almost impossible to perform reliably.<br><br>In short: the small superscript number is powerful precisely because it encodes the structure of repeated multiplication, and that structure obeys consistent, provable rules. Those rules do not have to be memorised as arbitrary facts — they can be derived from first principles, as you have done throughout this unit. That is what makes indices not just a shorthand, but a genuinely mathematical tool. |

## RUBRIC

| Criterion | Excellent (4) | Proficient (3) | Developing (1–2) |
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| Understanding of Index Notation and Its Purpose | Clearly explains that $a^n$ means repeated multiplication of base a exactly n times. Gives a compelling, accurate reason why index notation is needed, directly referencing the Kenya Census or another large/small quantity. Explanation is fully in the student's own words. | Correctly defines $a^n$ and explains the purpose of index notation with a relevant example. Minor gaps in depth or clarity but the core understanding is evident. | Shows partial understanding — may define the base or index correctly but not both, or gives a purpose without connecting it to a specific quantity. Example may be missing or incorrect. |
| Correct Application and Derivation of the Laws of Indices | States at least four laws (multiplication, division, power of a power, zero index, and/or negative/fractional index) correctly. For each, provides a clear mathematical derivation using expansion of repeated multiplication — not just a statement of the rule. All calculations are accurate. | States and correctly applies at least three laws. At least two include a derivation or explanation of why the law works. Calculations are mostly accurate with at most one minor arithmetic error. | States at least two laws but applies them with errors, or states them correctly without any explanation of why they hold. Derivations are absent or incomplete. |
| Use of Real-Life Kenyan Contexts | Includes at least three distinct, accurate real-life examples drawn from Kenyan contexts (e.g., census data, KEMRI research, tea farming in Kericho, Nakuru land measurement, CBK budget figures). Each example is directly linked to a specific law or concept rather than mentioned superficially. | Includes at least two Kenyan real-life examples that are accurately linked to index concepts. Examples are relevant but may lack depth of connection to the specific law being illustrated. | Includes one or fewer real-life examples, or examples are present but not accurately connected to a specific index law or concept. |
| Logical Explanation of the Zero and Negative Index | Proves $a^0 = 1$ using the division law (showing $a^n \div a^n = a^0$ AND = 1, therefore $a^0 = 1$). Explains $a^{-n} = 1/a^n$ by expanding a division where the denominator has more factors. Both explanations are logically sequenced and free from circular reasoning. | Correctly states and illustrates $a^0 = 1$ and $a^{-n} = 1/a^n$ with a numerical example. The logical link to the division law is present but may not be fully articulated as a formal derivation. | States that $a^0 = 1$ or $a^{-n} = 1/a^n$ without explanation, or provides an incorrect or circular justification. Numerical examples may be missing or inaccurate. |
| Quality and Completeness of the Final Driving Question Answer | The final answer directly addresses all four required elements: purpose of index notation, at least three laws with reasoning, at least two Kenyan contexts, and a thoughtful reflection on what would be lost without index notation. Writing is coherent, mathematically precise, and demonstrates genuine understanding rather than memorisation. | The final answer addresses at least three of the four required elements. The response is clear and mostly complete; reflection on the absence of index notation is present but may be brief or general. | The final answer addresses fewer than three required elements, or includes significant mathematical errors. The response reads as a list of recalled facts rather than a connected explanation. |